Not-ruled-out images of policy contrasts

A general procedure, with a planted village ABM as illustration

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This note writes the pipeline in generality first, then uses one implemented simulator as an example. The reporting object is a *not-ruled-out image* of a named policy contrast. That is the same *reporting logic* as a partially identified set—do not replace a set by a convenient point—not a claim that the object has the theorems of a Manski identified set.

Lectures in docs/lectures/ were drafted *before* waves. They are not this write-up. Numbers below come from the planted JSON in results/ and the archived F-empty file. Foil 1 on this pin is OLS-HC1, not a forest and not BCF. Overlay verdicts are written as set relations on subsets of $\mathbb{R}$ (Section 6).

1 The planner's problem

A planner compares assignment rules (policies) that act on a population of interfering units. Individual contrasts $Y_i(1) - Y_i(0)$ are not the target: under interference the argument of Y_i is the whole assignment vector. The natural estimand is a contrast of rules,

$$\theta(\pi, \pi'; c) \;=\; \mathbb{E}_c\big[\bar{Y}(\pi)\big] \;-\; \mathbb{E}_c\big[\bar{Y}(\pi')\big]\,, \tag{1}$$

where c indexes a complete data-generating process (a kernel, an ABM, a structural model) and the expectation is over whatever residual randomness the kernel still has once c is fixed.

The planner does not know c. What she has is a class of kernels $\{f(\cdot\,; c) : c \in C\}$ and a low-dimensional vector of moments z that she is willing to treat as the right functionals of the system. ("Willing" here is an identifying assumption about *which summaries to match*. It is not who named θ. Augusto froze θ_2 as a contrast of terminal means in METHODOLOGY.md §4.) History matching then *rules out* calibrations. It does not produce a posterior over worlds. The decision-relevant object is not the surviving set in parameter space. It is the image of θ over that set.

Two further incompletenesses sit beside calibration, and must not be folded into the same interval: (i) the scoring of the outcome Y (which operationalization?), and (ii) the fact that default estimators are one-world objects. The procedure below keeps the three sources on separate axes.

2 Objects

Kernel. A complete mapping f from (c, ω, π) to a panel (Y_T, D, X, A). Here ω is simulation noise, π is an assignment rule, A is a (possibly stochastic) network, X are covariates. The kernel is complete: incompleteness lives in c, not in unobserved counterfactuals inside f (Section 3). Implemented as vectorized algebra (kernel.py), not as a turtle-

by-turtle framework. NetLogo would be a twin of the same f, not a different science.

Box. $C \subset \mathbb{R}^d$ is the a priori range of free coordinates of c. Coordinates that are not up for calibration (in the example, η) are frozen inside f, not inside C.

Named estimands. A short list of contrasts, declared before any wave. Under interference the argument of Y_i is the whole assignment, so

$$\theta(\pi, \pi'; c) \;=\; \mathbb{E}_c\big[\bar{Y}_T(\pi)\big] \;-\; \mathbb{E}_c\big[\bar{Y}_T(\pi')\big]\,. \tag{2}$$

In the illustration the named list is

$$\theta_1(c) = \mathbb{E}_c[\bar{Y}_T \mid \pi_1] - \mathbb{E}_c[\bar{Y}_T \mid \pi_0], \tag{3}$$

$$\theta_2(c) = \mathbb{E}_c[\bar{Y}_T \mid \pi_2] - \mathbb{E}_c[\bar{Y}_T \mid \pi_0] \quad \text{(headline)}, \tag{4}$$

$$\psi(c) = \theta_2(c) - \theta_1(c) \quad \text{(targeting premium)}. \tag{5}$$

$\delta_{\text{spill}}(c)$ is a separate exposure-mapped spillover on the only arm that has a design (Section 5.7). Notation: the history-matching threshold is τ_{I}, not ψ. The two must not share a glyph. Expectations are over ω only (π_0 and π_2 are deterministic given A; π_1's extra randomness is part of ω).

θ is defined on the *terminal* village mean $\bar{Y}_T$, not on the 20-day path. That is how the functional was named, not a proof that path summaries would be wrong. A path functional would be a different freeze, and then M and z should be path summaries too.

Moment map. $M : C \to \mathbb{R}^k$ with $k < d$ by design. Underidentification in c is intended; the deliverable is a set in c-space. Whether θ moves on that set is empirical. M is not a menu of outcome scorings.

Anchors. $z \in \mathbb{R}^k$ is the target for M. In a simulation study z may be planted from a known $c^\star \in C$. In an empirical study z is external. Either way, z is frozen before waves.

Implausibility.

$$I(c) \;=\; \big(z - \widehat{M}(c)\big)^\top V(c)^{-1} \big(z - \widehat{M}(c)\big), \tag{6}$$

a squared Mahalanobis discrepancy, not a distance. $V(c) = \text{diag}\big(\sigma^2_{\text{samp}}(z) + \sigma^2_{\text{MC}}(\widehat{M}(c)) + \delta^{\circ 2}\big)$. The three pieces are sampling error in z, Monte Carlo error in the candidate moment, and an explicit model-discrepancy budget δ. V does not contain survey-measure noise.

Not-ruled-out-yet set. NROY $= \{c \in C : I(c) \le \tau_{\text{I}}\}$. Membership means we are not entitled to discard c. It does not mean c is observationally equivalent to the truth, and it is not a posterior.

Images.

$$\Theta_{\mathsf{hm}} = \{\theta_2(c) : c \in \mathsf{NROY}\}, \qquad \Theta_{\mathsf{exp}} = \{\delta_{\mathsf{spill}}(c) : c \in \mathsf{NROY}\}. \tag{7}$$

Reported as empirical $[L, U]$ over a pre-registered sample of survivors. No measure (uniform, posterior, …) is placed on NROY. Width is empirical: distinct survivors *can* map to different θ; they need not.

Diagnostic worlds. Two points $c_0^{\min I}$, $c_0^{\mathsf{med}\, I}$, functions of I only. Never of θ. "Best match" and "non-extremal" are the rationales; neither is theoretically privileged.

Measures. Scorings m of Y_T, frozen as a menu, admitted or excluded by a pre-declared restriction. $M^\star$ is a set of measures, not a subset of NROY. $\Theta_{\mathsf{cv}}(c_0)$ is the image of the estimand over $M^\star$ at that world.

Panel. One run of f is one micro-dataset

$$\mathcal{D}(c, \omega, \pi) \;=\; \big\{(Y_{iT},\, D_i,\, X_i,\, A_{i\cdot})\big\}_{i=1}^{N}. \tag{8}$$

World $\neq$ panel. One-world means the estimator is not given NROY; it conditions on one c. One panel is one ω. Kernel truth $\theta(c)$ is still one-world and uses $R = 200$ paired reps. Averaging a foil over 200 panels at the same c_0 would still be one-world.

Foils. One-world estimators fitted at each c_0 on *one* panel. Each foil is judged against *its own* image and against truth at that world (Section 6). They are not candidates for Θ_{hm}.

3 Potential outcomes, and what counts as data

Hahn / Rubin–Neyman with SUTVA: $Y_i = Y_i(D_i)$, $D_i \in \{0, 1\}$. One of two numbers per person is missing. That missingness *is* the incompleteness. BCF, an honest forest, or OLS-HC1 is a gold-standard interval for that SUTVA functional, given one dataset, at one world. The ruler, not the villain. Not "the forest failed under interference."

This class drops SUTVA ($\gamma > 0$, peer *outcomes*): $Y_i = Y_i(\mathbf{d})$, $\mathbf{d} \in \{0, 1\}^N$. The individual contrast $Y_i(1) - Y_i(0)$ is undefined unless everyone else's assignment is named. The well-defined object is a Hudgens–Halloran overall / policy contrast $\theta(\pi, \pi'; c)$ as above.

In the *field*, other $\bar{Y}(\pi)$ are still missing (one campaign). That would be Manski incompleteness in the potential outcomes. In the *simulator*, f is complete: run both arms, pair ω, subtract. Nothing inside f is an unobserved counterfactual. Remaining incompleteness is which $c \in C$ (five free knobs, three moments). $\Theta_{\mathsf{hm}} = \{\theta_2(c) : I(c) \le \tau_{\mathrm{I}}\}$ is a not-ruled-out image, not a Manski identified set and not a set of still-possible $Y_i(0)$. Same *reporting logic* (do not replace a set by a convenient point); not the same theorem.

$d > k$ underidentifies c. Width of Θ_{hm} is measured. Matching arm *levels* while θ_2 is a *contrast against control* makes unpinning possible, not a theorem (Section 5.5).

Only two objects have empirical analogs.

empirical analog	simulator object	who consumes it
one campaign's micro data	one panel $\mathcal{D}(c, \omega, \pi)$	OLS-HC1, Hájek
calibration targets	z (plus sampling variances, δ)	$I(c)$, NROY
survey operationalization	$s = m(Y_T)$	foils / Θ_{cv}

Not data: LHS points, moment and estimand Monte Carlo reps, NROY survivors, the image sample, $\theta(c)$. Those are probes of f. NROY points are arguments of $\theta(\cdot)$, not extra surveys.

4 Order of operations

The order is part of the identification statement. Using θ to decide which c survive is fishing.

1. Freeze f, C, z, δ, τ_{I}, the estimands, the measure menu and its admission rule, and the c_0 rule.
2. Draw the environment (A, X) once.
3. History-match: evaluate $I(c)$ on a space-filling design, keep $I \leq \tau_{\text{I}}$, optionally concentrate later waves on the surviving box. NROY is the union of survivors. Empty NROY is a finding: the pair (f, z) is rejected. Do not move τ_{I} or δ after seeing the waves.
4. Leg A. Sample survivors. Compute $\theta(c)$ and $\delta_{\text{spill}}(c)$ from the kernel. Report Θ_{hm} and Θ_{exp} as $[L, U]$.
5. Leg B. At each c_0, fit the one-world foils on a single panel. Compare each interval to its own image and to kernel truth at that c_0.
6. Measures. At those same c_0, score Y under each m, admit, report Θ_{cv}. Side branch, not a gate on legs A and B.

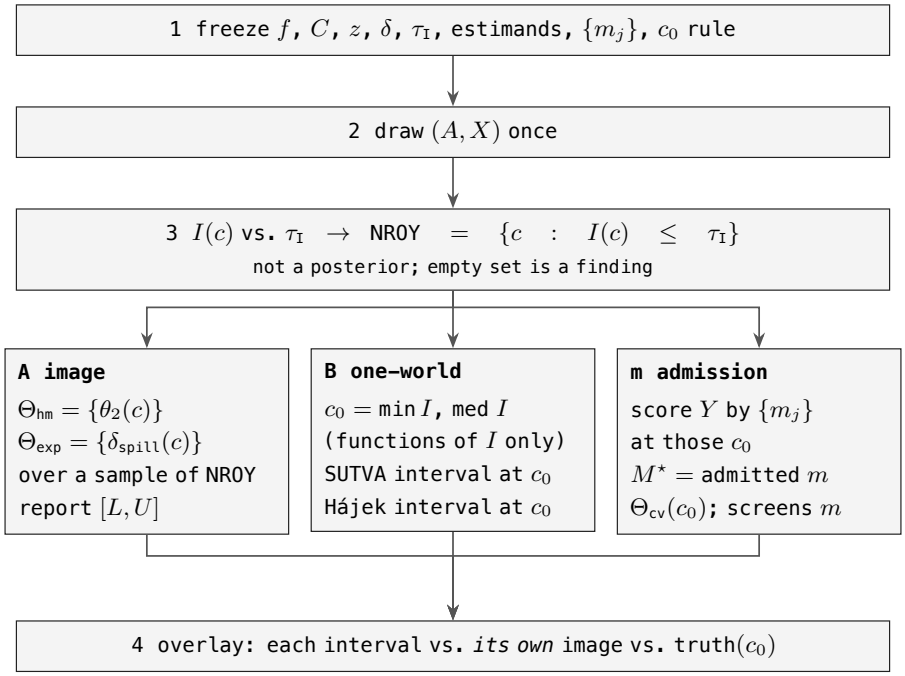


Figure 1: Order of operations. Legs A and B, and the measurement branch, are three reports. They do not merge into a single "construct-identified-robust" object.

5 Design of exogenous quantities

Everything below is frozen before waves. The justifications are design justifications, not data-dependent ones.

5.1 Threshold τ_{I}. $I(c)$ is a k-dimensional squared Gaussian form if $z - \widehat{M}$ were normal with covariance V. Taking $\tau_{\mathrm{I}} = F^{-1}_{\chi^2_k}(0.95)$ is the conventional history-matching cutoff (a 95% ellipsoid). It is a ruling-out convention, not a posterior credible region. Moving τ_{I} after seeing NROY would convert a pre-registered filter into a fit.

5.2 Discrepancy δ. Kennedy–O'Hagan discrepancy is the admission that even the right c need not hit z exactly, because f is not the world. On a correctly specified simulator δ is conservative: NROY is larger than a $\delta = 0$ match. Keeping a pre-declared δ anyway separates "the analyst's budget" from "what the DGP can hit." Loosening δ after a wave is the same sin as loosening τ_{I}.

5.3 The box C. Wide enough that underidentification is possible ($d > k$), narrow enough that every coordinate is economically signed (persistence, peer weight, and own-treatment shift positive; intercept not so free that Y saturates for every c). Excluding $\gamma = 0$ is a statement of subject-matter: the paper is about interference; a no-interference world surviving calibration would be its own finding and must not be slipped in.

5.4 How c is drawn. Nobody picks a favourite five-tuple by looking at θ. Wave 1 is a Latin Hypercube of 1000 points in C. Each candidate is kept iff $I(c) \leq \tau_{\mathrm{I}}$. Waves 2–3

redraw LHS in the survivors' bounding box, inflated 20% and clipped back to C. NROY is the *union* across waves. The plant $c^\star$ is the centre of C, chosen without looking at θ. Diagnostic c_0 are min I and median I on that union.

5.5 Anchors z. Two admissible sources, with different claims.

External z. Moments with a policy story and a kernel analog. If no $c \in C$ satisfies $I(c) \le \tau_{\mathrm{I}}$, that is F-empty: the pair (f, z) is rejected. That finding is about the analog, not about θ. Empty is the opposite of a wide Θ.

Planted z. $z := \widehat{M}(c^\star)$ at a $c^\star$ chosen without looking at θ. The illustration uses the centre of C. Planting makes NROY feasible by construction and makes θ_{true} known, which is the only place an overlay is checkable.

There are *three* moments, not five. Five is d, the free coordinates of C ($\alpha, \beta, \gamma, \kappa, \sigma_\varepsilon$). η is frozen inside f, not a moment and not in C. The three slots, in order:

slot (historical name)	village functional	arm
level_imm	$\mathbb{E}[\bar{Y}_T \mid \pi_2]$	hub level
gradient_edu	$\mathrm{corr}(Y_T, x_1 \mid \pi_2)$	literacy gradient
level_internet	$\mathbb{E}[\bar{Y}_T \mid \pi_1]$	random level

Control's mean $\mathbb{E}[\bar{Y}_T \mid \pi_0]$ is *not* in z. That is how z was named, not a theorem that it must be omitted. World Bank slots were three campaign-era / country aggregates (immunization, education gradient, internet); there is no untreated planet in that file. After F-empty, AMENDMENT-02 kept the same three functionals and planted them at $c^\star$. METHODOLOGY does not contain a paragraph "exclude π_0 so that θ_2 is set-identified." Underidentification of c is designed ($d = 5$, $k = 3$). Unpinning of θ_2 is a *possible* consequence of matching levels while θ_2 is a contrast against an unmatched control, plus δ. Width remains measured.

If control were added, $k = 4$ and c would still be a set, but θ_2 would sit close to a difference of two matched moments plus the δ slop. That would be a different freeze. Neither adding it to manufacture tightness nor dropping it to manufacture width is a scientific reason.

The illustration ran both sources. Country aggregates as z returned F-empty and were archived. The overlay uses the plant.

5.6 Diagnostic c_0. Min I mimics best-fit / MSM practice. Median I is a non-extremal rule on the same ranking. Both are functions of moment fit, not of the estimand. Two worlds, not four hundred: looping the estimator over every survivor answers "what if we gave the learner the calibration set," which default practice does not.

5.7 Foils. Foil 1 is a SUTVA one-world interval at c_0. The frozen role is that ruler: an honest interval *conditional on a world*, not a defendant charged with failing under interference. Preferred software is an honest causal forest; this pin does not load it (econml against numpy==2.4.6). The run therefore uses the declared fallback: OLS of the scored outcome on D given X, HC1 interval. Report the name. Do not call it BCF.

BCF remains an extra row only if asked for, same c_0, no new rule.

On one π_2 panel the regression is $s_i = \alpha + \tau D_i + X_i^\top \eta + u_i$. Stack $Z_i = (1, D_i, X_i') \in \mathbb{R}^k$ with $k = 5$ here (intercept, D, three covariates). OLS $\hat{\beta} = (Z^\top Z)^{-1} Z^\top s$, and $\hat{\tau}$ is the D coordinate. HC1 is the degrees-of-freedom sandwich

$$\widehat{\mathrm{Var}}_{\mathrm{HC1}}(\hat{\beta}) \;=\; \frac{n}{n-k}\,(Z^\top Z)^{-1}\Big(\sum_{i=1}^{n} \hat{u}_i^2\, Z_i Z_i^\top\Big)\,(Z^\top Z)^{-1}. \tag{9}$$

I_{OLS} is $\hat{\tau} \pm z_{1-\alpha/2}\,\mathrm{SE}(\hat{\tau})$ with SE the square root of the D diagonal. At $N = 800$, $n/(n-k) = 800/795 \approx 1.006$; the name is the convention, not a material correction. Rows are treated as uncorrelated. That is already a SUTVA-style assumption, not a network standard error. On the hub panel D *is* "being a hub," so $\hat{\tau}$ lives near δ_{dir} (hubs vs non-hubs inside one campaign), not near θ_2.

Foil 2 is design-based and exposure-mapped, and therefore lives only on an arm with a known assignment *lottery*. "Design" here is not "research design" and not "we designed an ABM." It means: can one write $p_i = \Pr_\pi(D_i = 1)$ and exposure probabilities from a known hat, before seeing Y?

π_1 is SRSWOR: simple random sample without replacement. Seed exactly 80 of 800 households; every 80-subset equally likely; no repeats. Inclusion $p_i = 80/800 = 1/10$. Neighbour counts among the untreated are hypergeometric, which is how $p_{i,S}$ are written without touching (α, β, γ). π_2 (top-80 degree) has no hat: given A, $p_i \in \{0, 1\}$. Inverse-probability weights are undefined. That un-reachability is reported, not walked around.

"Estimable from one designed panel" means: from one realized randomized campaign plus the known lottery, form $\widehat{\delta}_{\mathrm{spill}}$ and $\widehat{\delta}_{\mathrm{dir}}$. It does *not* mean θ_2 is identified from one campaign. θ_2 needs π_2 and π_0. Even a field hub-campaign does not see control, and was not a lottery.

Exposure $g_i = (D_i, s_i)$ with the same k_i as in the kernel (degree, isolates patched):

$$k_i = \max\Big(\sum_j A_{ij},\, 1\Big), \qquad s_i = \frac{\sum_{j:A_{ij}=1} D_j}{k_i}. \tag{10}$$

s_i is the share of neighbours' *treatment*, not neighbours' outcomes. Isolates have $k_i = 1$ and $s_i = 0$. Path $A{-}B{-}C{-}D$, treat $\{B\}$: $s = (1, 0, 0.5, 0)$. Strata at cut 0.25 are the sets

$$\begin{aligned} S_1 &= \{i : D_i = 1\}, \\ S_{0a} &= \{i : D_i = 0,\ s_i = 0\}, \\ S_{0b} &= \{i : D_i = 0,\ 0 < s_i < 0.25\}, \\ S_{0c} &= \{i : D_i = 0,\ s_i \geq 0.25\}. \end{aligned}$$

On that path: $A, C \in S_{0c}$, $B \in S_1$, $D \in S_{0a}$, $S_{0b} = \emptyset$. If any cell has fewer than 20 agents in a replication, S_{0b} is merged into S_{0c} for all reps of that world.

Kernel truth uses unweighted cell means of oracle Y, then averages over random-arm reps:

$$\delta_{\texttt{dir}}(c) = \mu_{S_1} - \mu_{S_{0a}}, \qquad \delta_{\texttt{spill}}(c) = \mu_{S_{0c}} - \mu_{S_{0a}}. \tag{11}$$

Foil 2 replaces μ_S by the Hájek mean on one π_1 panel,

$$\widehat{\mu}_S \;=\; \frac{\sum_i \mathbf{1}\{i \in S\}\, Y_i / p_{i,S}}{\sum_i \mathbf{1}\{i \in S\}/p_{i,S}}. \tag{12}$$

Glue the two μ_S and the oracle gate is unreadable. The exposure map is neighbour-D; the kernel peer term is neighbour-Y over T ticks (a Sävje-style misspecification of the map). That is a limitation of Foil 2, not a license to replace it with a second SUTVA learner.

Foil 2 does *not* estimate θ_2. Two different one-world referents, neither of which is the calibrated-simulator report: Hahn classroom $\rightarrow$ Foil 1; a network RCT that admits interference and uses an exposure map (Aronow–Samii family) $\rightarrow$ Foil 2. The goal of the second row is an interference-aware interval that (i) can write spillover, (ii) is identified by a lottery from one panel, (iii) still does not consume NROY, so I_{E2} versus Θ_{exp} isolates calibration after architecture (Section 6). Rank replacements by that goal. Dropping Foil 2, or swapping it for BCF, returns the architecture/calibration confound. Stochasticizing hub-seeding so IPW can hit it is a different policy. Θ_{exp} is the image of the *kernel* $\delta_{\texttt{spill}}(c)$, not of the Hájek interval.

5.8 Measures. A short menu, including a positive control (oracle $s = Y$) and a scoring designed to fail a restriction stated in English before the wave ("must not load on a treatment-channel covariate"). Admission is mechanical. It does not look at θ.

5.9 Monte Carlo budgets. Moment evaluations use fewer replications than estimand evaluations: ruling-out has to be cheap enough to search C; $\theta(c)$ is computed only on survivors. The plant uses the estimand budget so that z is not noisier than the object we will later quote.

6 Interval geometry

Sampling intervals and images are subsets of $\mathbb{R}$. Verdicts are inclusions, not slogans.

Let $I = [\iota_L, \iota_U] \subset \mathbb{R}$ be a closed sampling interval (a foil CI at one c_0) and let $\Theta = [L, U] \subset \mathbb{R}$ be the reported image of the *same* functional over a pre-registered sample of NROY (empirical min/max; not a continuum and not a posterior). Write $\text{int}(\Theta) = (L, U)$.

Implemented labels (METHODOLOGY §10; `classify_interval`):

$$\texttt{F-HONEST} \iff \Theta \subseteq I, \tag{13}$$

$$\texttt{F-OVERCONF} \iff I \subset \text{int}(\Theta). \tag{14}$$

Anything else is unnamed — no third finding after the freeze — including

$$I \cap \Theta = \emptyset \quad \text{and} \quad I \cap \Theta \neq \emptyset \text{ with neither inclusion.} \tag{15}$$

The code does not additionally test whether θ_{true} sits outside I. METHODOLOGY's prose "and excludes θ_{true} or parts of $[L, U]$" is already implied by $I \subset \text{int}(\Theta)$ for the image endpoints; $\theta_{\text{true}}(c_0)$ is checked separately by the oracle gate.

Oracle gate (harness, not a finding):

$$\theta_{\text{true}}(c_0) \in I \quad \text{or, for Foil 2,} \quad \delta_{\text{spill}}(c_0) \in I_{E2}. \tag{16}$$

The point estimate is unused. Fail $\Rightarrow$ stop; do not read overlay stickers.

Each foil versus *its own* image:

$$(I_{\text{OLS}},\ \Theta_{\text{hm}}), \qquad (I_{E2},\ \Theta_{\text{exp}}). \tag{17}$$

Cross-functional geometry (I_{OLS} against Θ_{hm} when OLS is not θ_2) is reported as a set relation, not forced into F-$*$.

Two gaps must not be dumped into one interval. Write

$$\mathcal{A} = \text{architecture (SUTVA vs exposure map vs kernel)},$$
$$\mathcal{C} = \text{calibration (which } c \in \text{NROY)}.$$

I_{OLS} versus Θ_{hm} can mix $\mathcal{A}$ and $\mathcal{C}$ when the functionals differ. Foil 2 is the attempt to handle $\mathcal{A}$ first: it can write spillover and still does not see NROY. Then, in order,

1. $\delta_{\text{spill}}(c_0) \in I_{E2}$ — architecture and sampling OK *at the world*;
2. $I_{E2} \subset \text{int}(\Theta_{\text{exp}})$ — residual width is variation of the *same* functional across surviving c.

That pair is what "I_{E2} versus Θ_{exp} isolates calibration after architecture" means. The leftover set $\Theta_{\text{exp}} \setminus I_{E2}$ is not a sampling CI and is not a SUTVA misspecification residual. F-OVERCONF-E2 is only the sticker for (2). It is almost automatic once (1) holds and the length of Θ_{exp} exceeds sampling noise. Empty or covering images are also allowed findings; the interesting object is the width, not the label.

7 Illustration: a planted village

One large village, $N = 800$ households, acquaintance network $G(N, 4/N)$ (mean degree ≈ 4). $Y_{i,t} \in (0, 1)$ is preventive-practice intensity—a score, not a country coverage rate. Covariates: literacy x_1, resources x_2, baseline propensity x_3. Three rules seed 80 households from $t = 0$: none (π_0); SRSWOR (π_1, the only arm with a design); the 80 highest-degree households (π_2, deterministic given A).

The kernel is linear-in-means with logistic saturation. For $t = 0, \ldots, T-1$ with $T = 20$,

$$Y_i(t+1) \;=\; \Lambda\Big(\alpha + \beta Y_i(t) + \gamma \frac{\sum_{j \in N(i)} Y_j(t)}{k_i} + \eta \cdot X_i + \kappa D_i + \varepsilon_i(t)\Big), \tag{18}$$

$\Lambda(z) = 1/(1+e^{-z})$, $\varepsilon_i(t) \sim N(0, \sigma_\varepsilon^2)$. Peers enter as neighbours' *lagged practice*, not as neighbour treatment. The five free coordinates and the frozen η are

	box / value	role
α	$[-2.5, -0.5]$	intercept of the index
β	$[0.05, 0.60]$	own-lag persistence
γ	$[0.05, 0.80]$	peer weight; 0 excluded
κ	$[0.25, 2.50]$	own-treatment shift (same κ on every arm)
σ_ε	$[0.01, 0.40]$	sd of ε, not a standard error
η	$(0.05, 0.10, 0.02)$	literacy, resources, propensity; not calibrated

$c^\star$ is the centre of C: $(-1.5,\ 0.325,\ 0.425,\ 1.375,\ 0.205)$. Frozen $z = \widehat{M}(c^\star) = (0.264,\ 0.050,\ 0.262)$ – hub-arm mean Y_T, corr(Y_T, x_1) on the hub arm, random-arm mean Y_T. The control mean is *not* in z, so θ_2 (a contrast against control) is not a moment being matched. $\delta = (0.02, 0.05, 0.02)$ and $\tau_{\mathrm{I}} = F^{-1}_{\chi^2_3}(0.95) \approx 7.81$ are the same numbers used in the rejected country-total attempt. The graph A is drawn once and not calibrated. Connectivity is not the latent being measured. The measurement layer scores Y, not edges.

8 PoC results

8.1 External z failed. Matching the same kernel to rescaled World Bank country totals (DPT coverage, education–immunization correlation, internet users; fetched 2026-08-12) ruled out every c. Wave 1 kept **0** of **1000**. Minimum $I = 204$ against $\tau_{\mathrm{I}} \approx 7.81$. Closest M in the box is still far from z (Figure 2). τ_{I} and δ were not moved. That finding stands for the pair (kernel, country aggregates). It is not a wide Θ. It is a reason to stop asking this kernel to match functionals it cannot speak to.

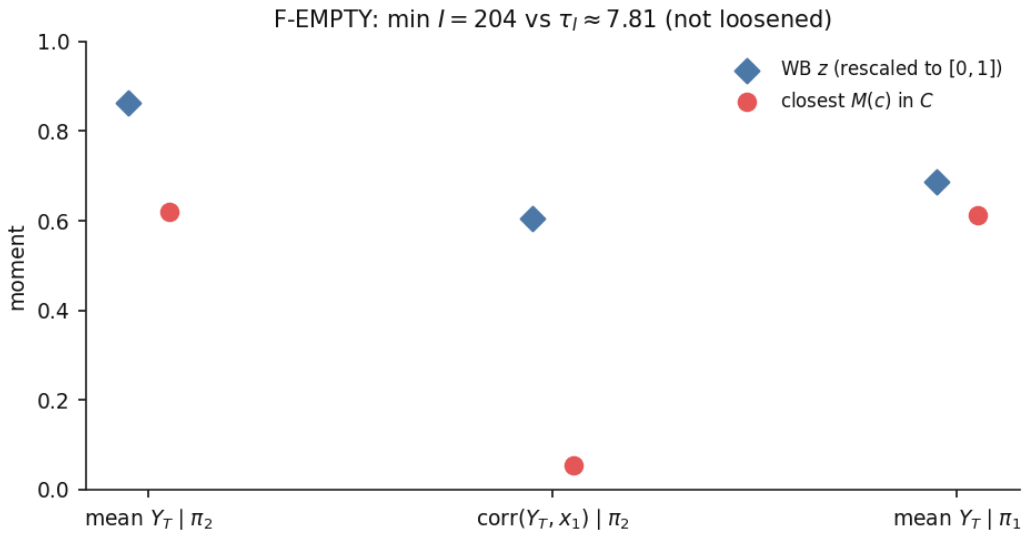


Figure 2: F-EMPTY on World Bank z (rescaled to $[0,1]$). Closest simulated moments in C versus anchors. Not a calibrated world.

8.2 Planted z left a fat NROY. Against $z := \widehat{M}(c^\star)$, three waves kept 202, then 416, then 373; union $|\text{NROY}| = 991$ (Figure 3). The set did not collapse. Closest survivor has $I \approx 0.002$. That is a close score, not a recovery of $c^\star$: several coordinates, β and γ among them, are not the plant values. That point is $c_0^{\min I}$ by the pre-declared rule, not because it won on θ. Median-I c_0 has $I \approx 2.02$ and a much larger κ.

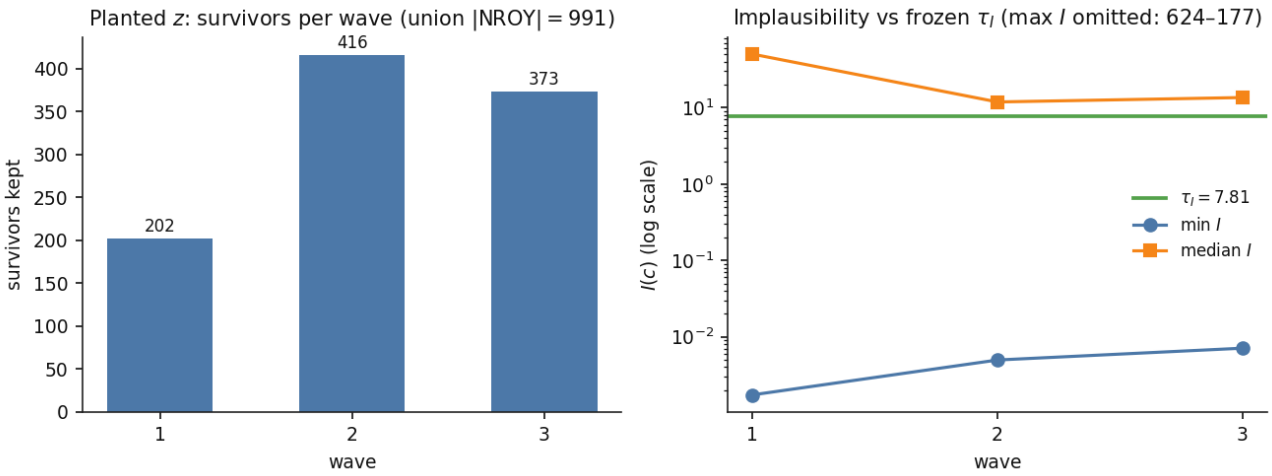


Figure 3: Planted history matching. Left: survivors per wave. Right: min and median $I(c)$ against frozen τ_I (log scale). Later waves concentrate; τ_I is not moved.

8.3 The image of θ_2 is not a point. At $c^\star$, hub vs none is $\theta_2 = 0.038$ (3.8 pp). Random vs none is $\theta_1 = 0.036$. Targeting premium $\psi \approx 0.002$. In this DGP, picking influencers barely beats picking at random. That is a fact about this $c^\star$, not a theorem about targeting.

Among 400 sampled survivors, θ_2 runs from 0.6 pp to 7.3 pp. The planted 3.8 sits inside. Honest report: $\Theta_{\text{hm}} = [0.006, 0.073]$, not 0.038. Not a sampling distribution and not a posterior.

Spillover $\delta_{\text{spill}}(c^\star) \approx 0.008$; image $\Theta_{\text{exp}} \approx [-0.001, 0.027]$. Direct-effect analogue (hubs vs non-hubs *inside* the random campaign) $\delta_{\text{dir}}(c^\star) \approx 0.34$.

object	min I world	median I world
$I(c_0)$	0.002	2.02
$\theta_2(c_0)$	0.037	0.063
OLS-HC1 I_{OLS} (hub panel)	$[0.326, 0.346]$	$[0.583, 0.595]$
I_{OLS} vs Θ_{hm}	$I_{\text{OLS}} \cap \Theta_{\text{hm}} = \emptyset$	$I_{\text{OLS}} \cap \Theta_{\text{hm}} = \emptyset$
$\delta_{\text{spill}}(c_0)$	0.006	0.008
Hájek I_{E2}	$[0.003, 0.009]$	$[0.006, 0.011]$
I_{E2} vs Θ_{exp}	$I_{E2} \subset \text{int}(\Theta_{\text{exp}})$	$I_{E2} \subset \text{int}(\Theta_{\text{exp}})$
oracle gate	$\delta_{\text{spill}}(c_0) \in I_{E2}$	$\delta_{\text{spill}}(c_0) \in I_{E2}$
Θ_{cv}	$[0.030, 0.037]$	$[0.056, 0.063]$
admitted m	m_1, m_2	m_1, m_2

8.4 Overlay. Figure 4 is the money plot (two c_0, two functionals). Figure 5 repeats the same numbers on a shared axis so the functional mismatch is visible without faceting.

At both c_0, $\delta_{\text{spill}}(c_0) \in I_{E2}$ (oracle gate: harness not broken) and $I_{E2} \subset \text{int}(\Theta_{\text{exp}})$. Architecture handled; residual is which surviving c. The sticker F-OVERCONF-E2 is that second inclusion.

OLS-HC1 on the hub panel lives near 0.33–0.59, the neighbourhood of δ_{dir}, not of θ_2. D *is* "being a hub." The geometry is $I_{\text{OLS}} \cap \Theta_{\text{hm}} = \emptyset$, which is unnamed (neither $\Theta_{\text{hm}} \subseteq I_{\text{OLS}}$ nor $I_{\text{OLS}} \subset \text{int}(\Theta_{\text{hm}})$). Not "OLS failed under interference."

Oracle and noisy-survey scorings of Y are admitted; the item that loads on baseline propensity is excluded, as designed. Θ_{cv} is a small scoring gap at a world, not a second NROY.

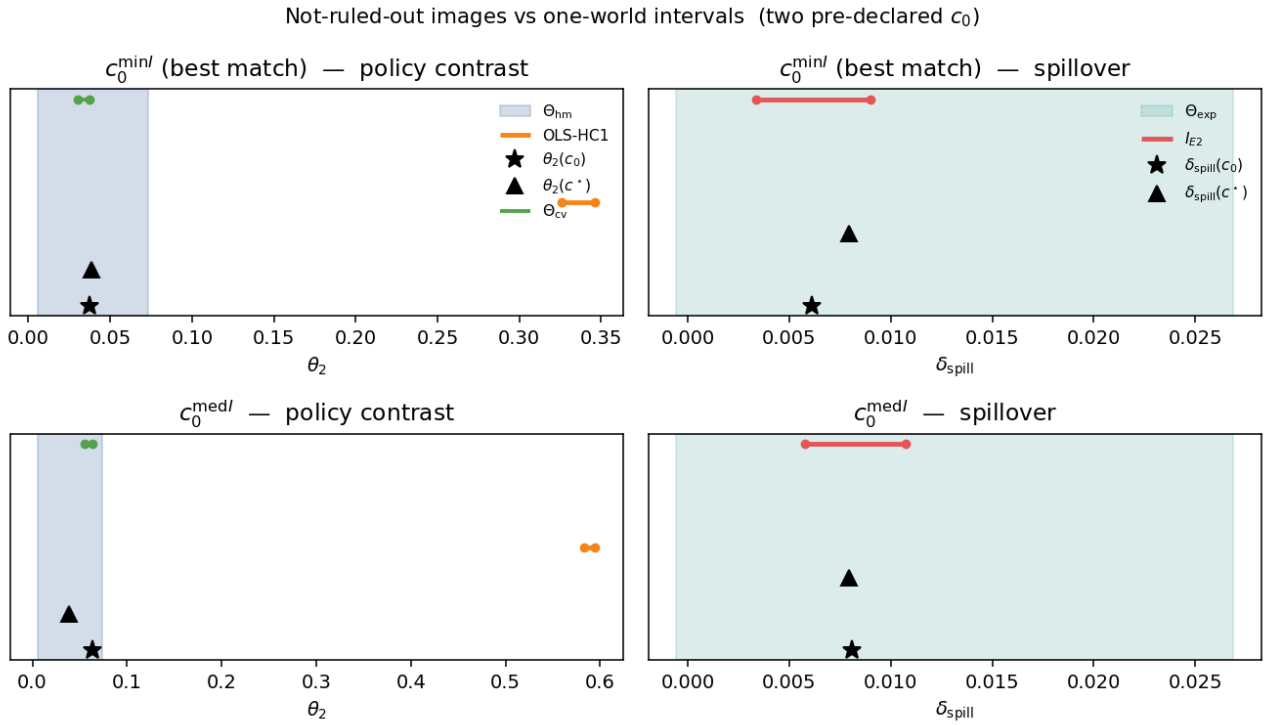


Figure 4: Not-ruled-out images versus one-world intervals at two pre-declared c_0. Left: θ_2 against Θ_{hm}; orange is OLS-HC1 (not a forest). Right: δ_{spill} against Θ_{exp}; red is Hájek. Each interval against *its own* image.

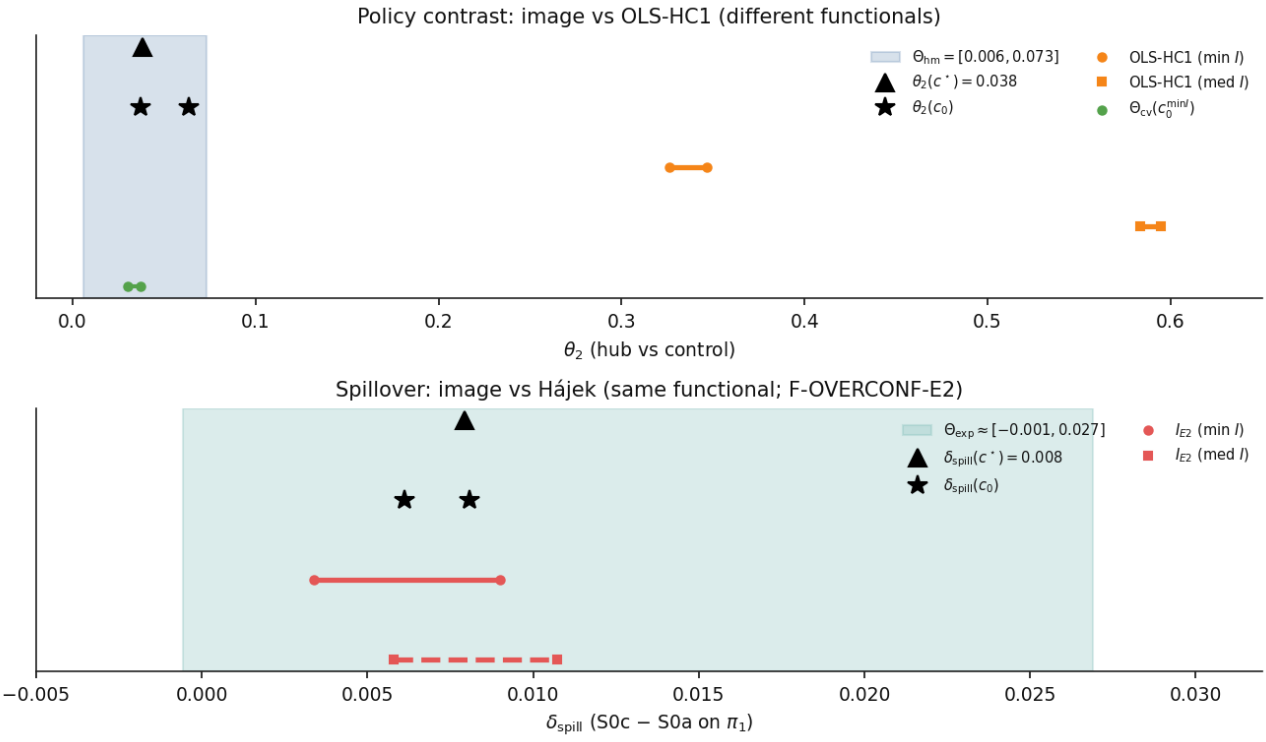


Figure 5: Same objects on one axis per functional. Top: $I_{\text{OLS}} \cap \Theta_{\text{hm}} = \emptyset$ because OLS is not θ_2. Bottom: $\delta_{\text{spill}}(c_0) \in I_{E2}$ and $I_{E2} \subset \text{int}(\Theta_{\text{exp}})$.

9 Who should care — and who should not

Slogan-level novelty is weak. History matching already produces NROY in input space (Craig, Goldstein, Vernon). Equifinality of ABMs is not a discovery of this note (Beven; Williams–Guikema–Brown–Agrawal map ensembles through policy). Partial identification already says: report the set. Interference already has overall vs direct (Halloran, Hudgens–Halloran). Honest forests and BCF already give intervals *given a panel*. None of those pieces is new. The remaining wedge, if this is ever a paper, is the *composition*: a named policy contrast under interference, as the image of a history-matched set, plus a measure-admission set, plus a paired SUTVA ruler and an interference-aware estimator that isolates calibration as the residual, on a simulator where truth is known. That is enough for an M.S. demonstration. It is not an Econometrica gap.

Anyone (method). Interesting if you already publish a policy number from a *calibrated simulator* (ABM, structural model, microsimulation) and currently report Monte Carlo error at a staff c_0. The overlay is a reporting split: sampling at a world is not uncertainty about which worlds z still allows. Empty NROY is a usable negative result (the analog is rejected; do not loosen the cutoff). Width of Θ is measured, not assumed: here θ_2 moves by about 7 pp across survivors, not by "anything."

Causal inference, empirical. Not a new identifying strategy for the designs STP 517 actually teaches. This procedure does not replace unconfoundedness, an instrument, DiD, or an RDD. It does not estimate a CATE $\tau(x)$ on a network. History matching does not identify heterogeneous treatment effects.

It *is* relevant to empirical CI in two narrower cases.

(1) *You are already using a model as the identification vehicle* — a structural DGP, an epidemic simulator, an ABM of a campaign — and then quoting one $\hat{\theta}$. Then the honest object is the image of that θ over calibrations not ruled out by the moments you were willing to match. One-world intervals (regression, forests, BCF) remain honest *at a world*. They are not the image. If your paper is IV for wages, this is not your appendix.

(2) *Interference is in the DGP you wrote, and ITE is undefined.* The well-posed object is a contrast of assignment rules. A SUTVA learner on a targeting arm, where D is a function of the graph, identifies something in the neighbourhood of "being a hub," not the campaign contrast. That is an identification remark, not an estimator bake-off. It is closer to STP 517 (name the estimand; when is regression causal) than to "run BCF on our panel."

This PoC, specifically. The overlay is checkable because z was *planted*. The one attempt at external z (World Bank country totals) returned F-empty. So this is not an

empirical estimate of immunization policy, and it should not be narrated as one. As a demonstration that the composition can be run, with pre-registered verdicts including F-empty and "neither F-$*$", it is a completed lab. As a contribution to empirical causal inference on real networks or real campaigns, it is not more than that lab — a toy with a clean object dictionary — unless someone later freezes a z this kernel can actually speak to, without moving τ_{I}.

If the audience is "applied micro who run regressions," treating this as a portfolio piece that shows you can name functionals and not cheat a cutoff is the honest pitch. If the audience already calibrates simulators for policy, the overlay is the thing to steal; the village story is not.

10 What the procedure is not

- Not a posterior over c, and not a distribution of treatment effects.
- Not a Manski identified set theorem-for-theorem. Same reporting logic; different incompleteness (calibration of a complete kernel, not missing counterfactuals); sampling and discrepancy already folded into V.
- Not a claim that equifinality of θ is guaranteed. Width is measured, not assumed.
- Not a finding that one-world estimators are bad at networks. They are honest for a world. The overlay asks whether one calibrated world is enough.
- Not a measurement model of the network. A is frozen.
- Not an empirical immunization study. WB z was rejected.

Implementation. ecn593-poc-abm, AMENDMENT-02. $|\mathsf{NROY}| = 991$, $\Theta_{\mathsf{hm}} = [0.006, 0.073]$, $\theta_2(c^\star) = 0.038$. Foil 1 estimator: OLS-HC1 (CausalForestDML did not import; not BCF). Overlay: figures/overlay.png. WB attempt archived as F-empty. The declared exposure cut at 0.5 is reported in the study packet (ecn593-applied-project, section "Declared cut at 0.5"), not here. This note does not repeat those digits. τ_{I} and δ were not moved for it. Figures in this note are redrawn from the headline JSON; those waves were not re-run.

A Tamer (2010) and this image

This appendix is a written object map, not a new finding and not a change to the freeze. It exists so a later thesis page, and an optional four-minute extension after a seminar on Tamer, can use one text. The seminar itself should teach Tamer's examples. Missing binary outcomes and the 2×2 entry game are the right objects for an audience meeting partial identification for the first time. The village is a bad teaching example for his theorems: it is a different incompleteness, and leading with it invites the collapse this note already refuses.

A.1 His objects. Tamer's review treats identification as a menu, not a yes/no bit. Assumptions plus the population distribution map to information about a parameter. More

assumptions buy a smaller set and less credibility; he opens on Manski's law of decreasing credibility. The identified set is the set of parameter values consistent with the maintained assumptions. When that set is the argmin of an objective Q, a sample analogue is a level set $\{y : Q_n(y) \leq c_n\}$. He separates two targets (his §4.1): inference on the whole set, and inference on a true parameter $y^\star$ known only to lie in the set. Under misspecification he notes there may be no true parameter in the set at all — only values that minimize a distance between the model class and the distribution of the data (his footnote 14). His worked examples are missing data (one of two potential outcomes is unobserved) and an entry game (multiple equilibria). Neither is a calibrated simulator.

A.2 Same reporting logic. Do not replace a set by a convenient point. That sentence is the entire overlap, and it is already the first paragraph of this note. The map below uses only objects this note has already reported. It does not import a new endpoint.

Tamer	This note	Stop
Assumption menu	Frozen box C, moments z, discrepancy δ, cutoff τ_{I}, named θ_2	Changing τ_{I} after seeing the set is changing the menu.
Level set of an objective	$\{c : I(c) \leq \tau_{\mathrm{I}}\}$. Planted run: $\lvert\mathsf{NROY}\rvert = 991$.	I is one joint ellipsoidal discrepancy, not a vector of moment inequalities. τ_{I} is a pre-registered cutoff. It is not a $c_n \to 0$ tuning sequence, and NROY is not claimed to be a Hausdorff-consistent estimator of an identified set.
True parameter inside the set	Planted $\theta_2(c^\star) = 0.038$ sits in $\Theta_{\mathsf{hm}} = [0.006, 0.073]$. The report is the interval.	The interval is an image of a functional over a sample of survivors ($S = 400$), not the identified set of c, and not a posterior on worlds.
Misspecified distance set, no $y^\star$ inside	World Bank z: wave 1 kept 0 of 1000; minimum $I = 204$ against $\tau_{\mathrm{I}} \approx 7.81$.	Empty is a finding about the pair (f, z). It is not a wide Θ, and it is not a warrant to loosen τ_{I} or δ.
Stronger assumptions, tighter claim	OLS-HC1 at the two c_0 is a few hundredths wide and disjoint from Θ_{hm}.	That interval conditions on one world and on a SUTVA regression of hub status. Disjoint means a different functional, not "the estimator failed."

The credible tightening in this design is not "add a parametric assumption and shrink Θ_{hm}." It is to keep three incompletenesses on separate axes, which the body of the note already does: which c is not ruled out; which scoring of Y is admitted at a world; which one-world interval is being quoted. Folding them into one Manski box would buy a single picture by spending the distinction.

A.3 Where the analogy stops.

- Incompleteness here is which complete kernel still matches z. Inside f, both policies can be run. Missing $Y_i(0)$ is Tamer's missing-data example. It is not this image.
- Θ_{hm} is a not-ruled-out *image* of θ_2. Tamer's identified set is a set of parameter values. Stopping at NROY would be stopping in his parameter space. The decision object is the image. Same reporting status; not his theorem.
- Membership in NROY means not ruled out yet by $I(c) \leq \tau_{\text{I}}$. It does not mean observational equivalence, and it does not mean the point is known to be true.
- V already folds sampling variance and the discrepancy budget δ into the score that rules c out. The computed image therefore mixes calibration uncertainty with those allowances. It is not a population identified set under exact moment equalities, and it is not a confidence set for one.
- Nothing in the results file is a Chernozhukov–Hong–Tamer region, a subsampled confidence set, or a uniformly valid interval for a partially identified parameter. Sampling uncertainty at one c_0 is the foil interval. Width of Θ_{hm} is a different axis (Section 6).

A.4 If the seminar gets four extra minutes. Speak this, then stop. Do not show the overlay unless asked.

> A simulator we wrote is a different incompleteness. Both policies can be run, so the missing counterfactual in the missing-data example is not what is unidentified. What the moments do not pin is the calibration. The level set of the implausibility score is a set of worlds. The object worth reporting is the image of a named policy contrast over that set. In the planted run that image was about 7 percentage points wide, and a tight regression interval at one surviving world missed it because the regression was not that contrast. Same discipline as the review: do not replace the set by a convenient point. Not the same theorem.

Do not say that Θ_{hm} is a Manski identified set, that NROY points are observationally valid, that $I(c)$ is a moment-inequality model, or that the seminar has just seen Tamer's inference section implemented. Those sentences are false of the finished artifacts, and they are the ones a careful listener will ask about.